

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 128

Model Hyperparameters:

- min_units: 128
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

LSTM Multi-Step Training Report

Report Generated: 2025-04-15 10:31:34

Training Start: 2025-04-15_09-23-51

Training End: 2025-04-15_10-02-26

Training Duration: 38.57 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: elu

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 1.8390

MSE: 7.1661

RMSE: 2.6770

R2 Score: 0.8373

Multi-Step Forecast:

Forecast Steps: 10

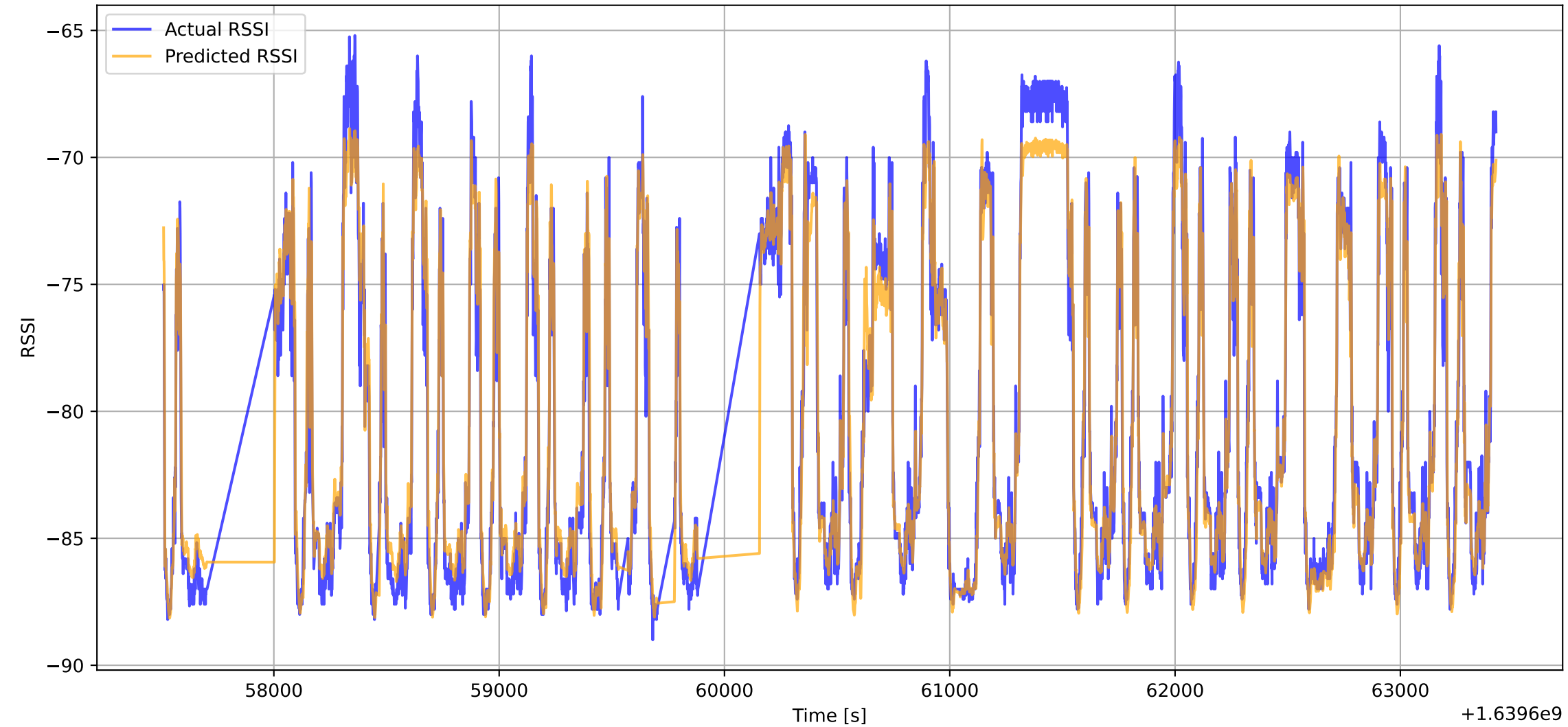
MAE: 0.0308

MSE: 0.0018

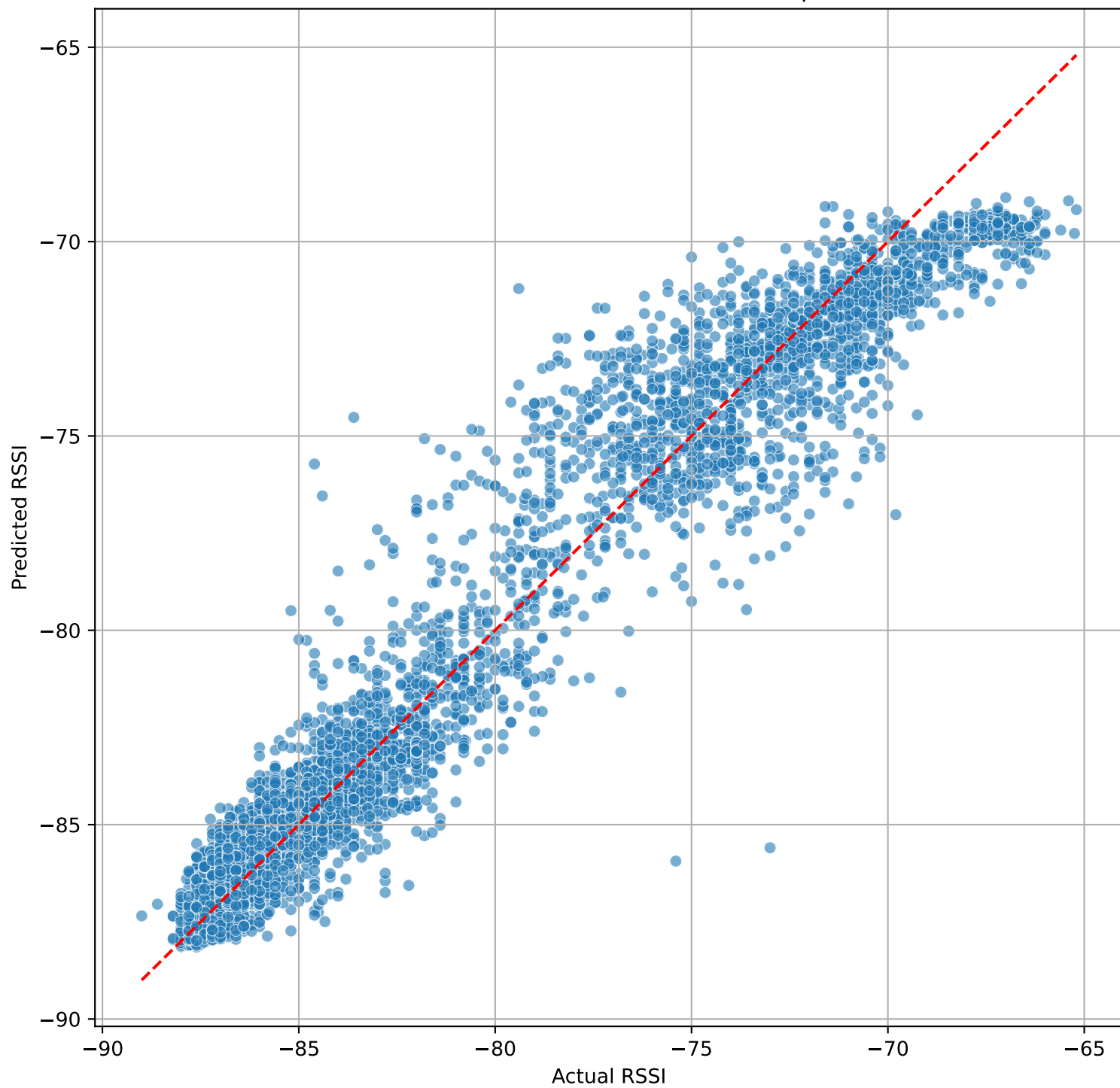
RMSE: 0.0424

R2 Score: 0.9474

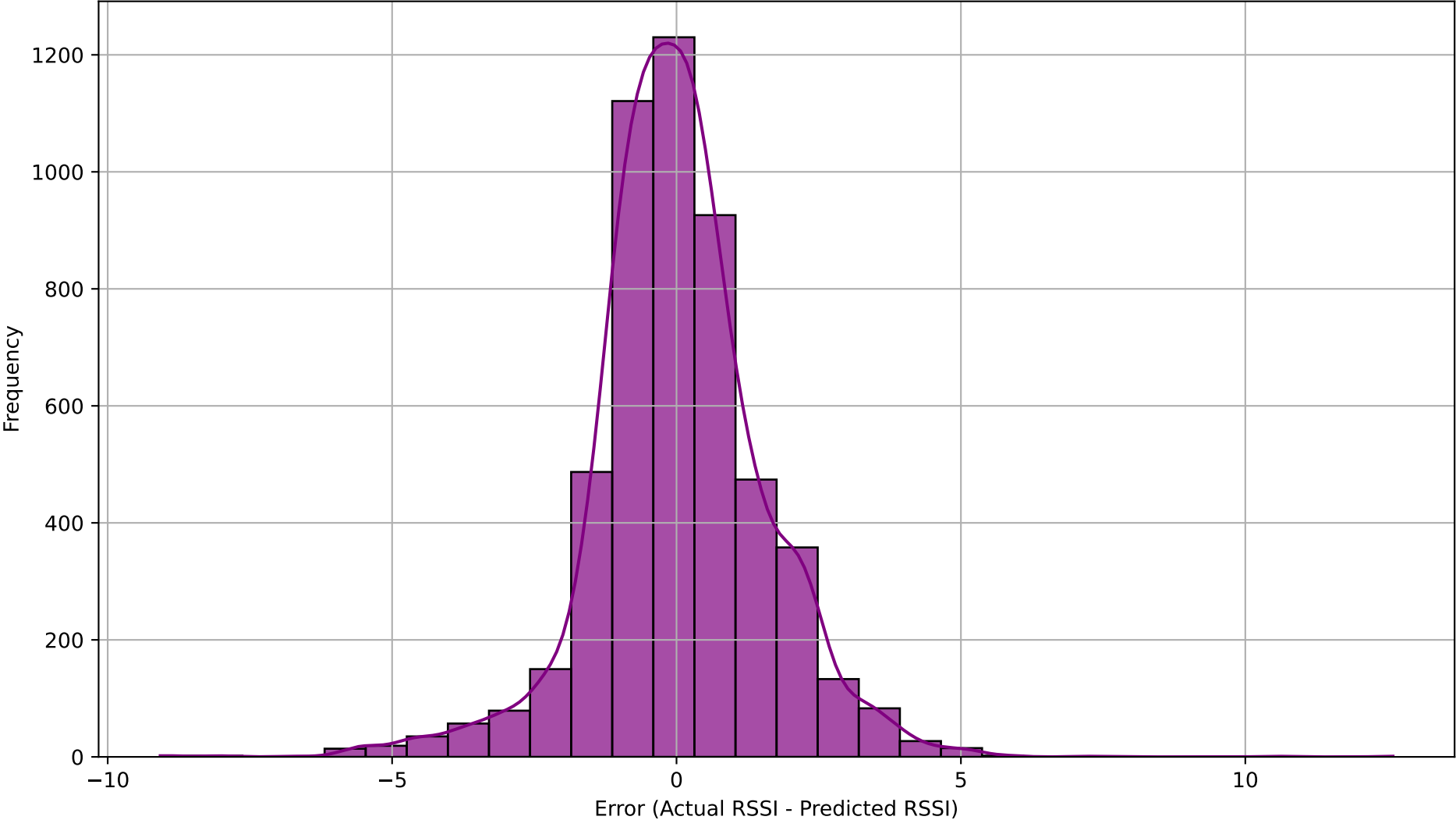
Actual vs Predicted RSSI (First Step)



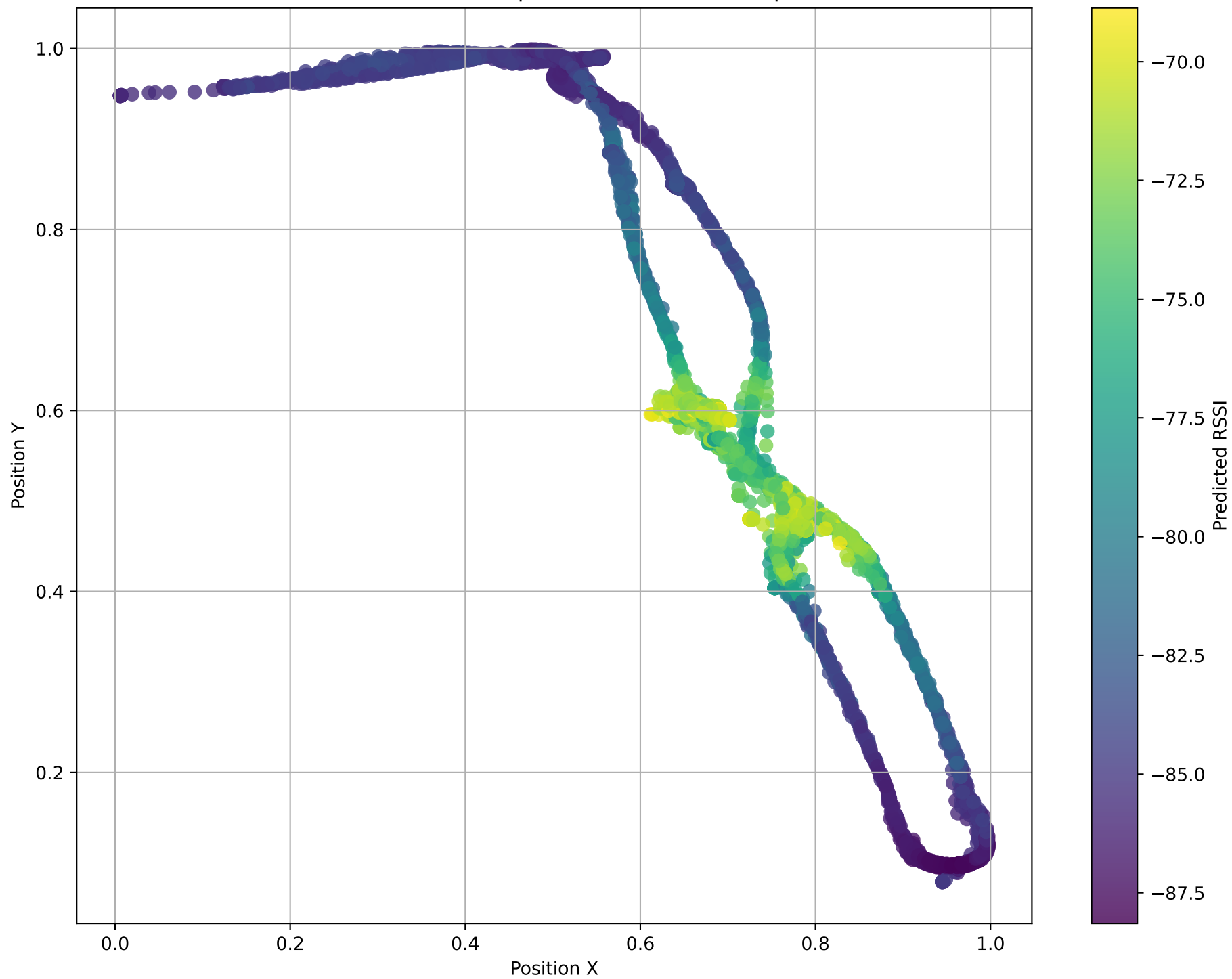
Actual vs Predicted RSSI (First Step)



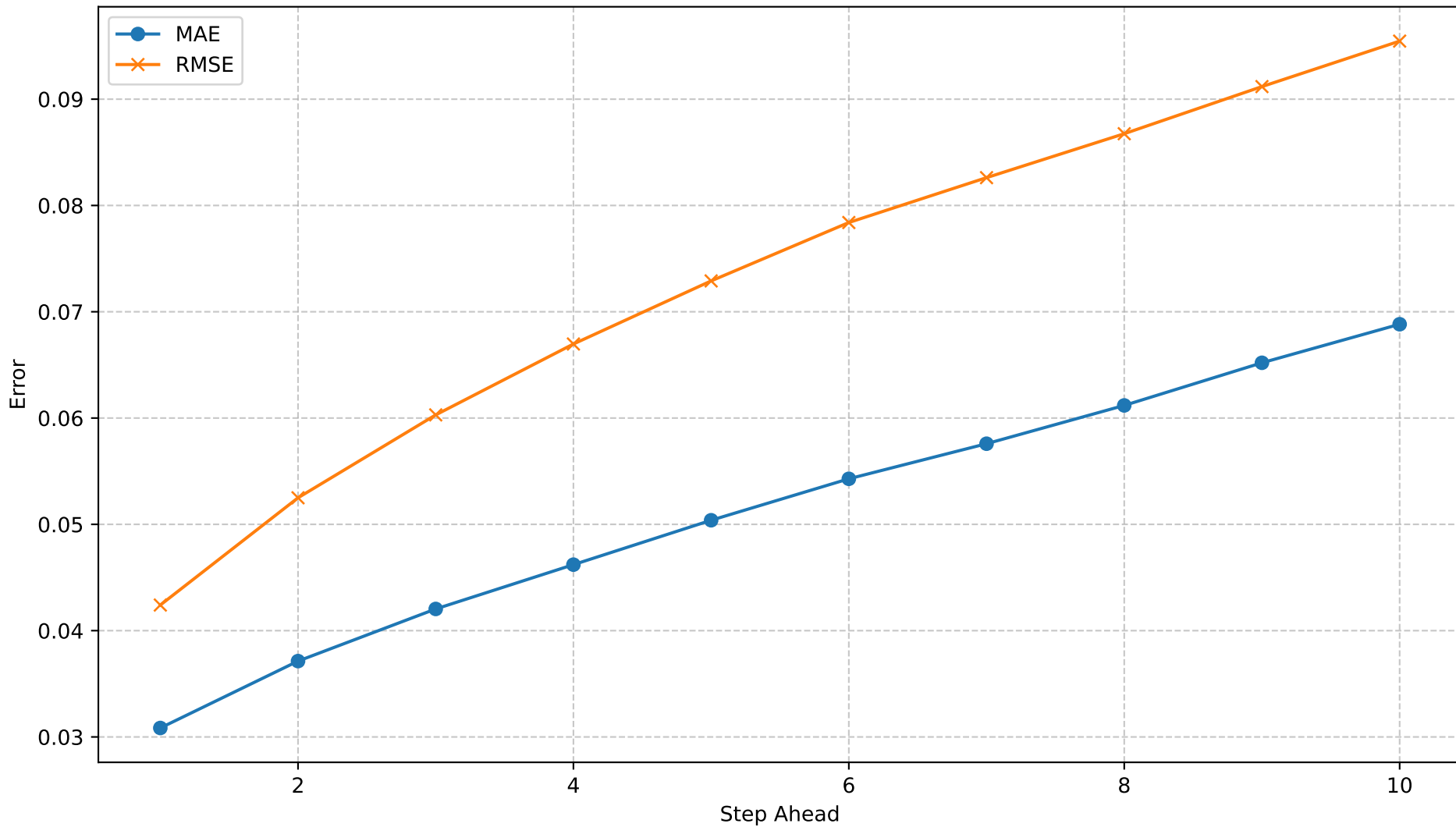
Distribution of Prediction Errors (First Step)



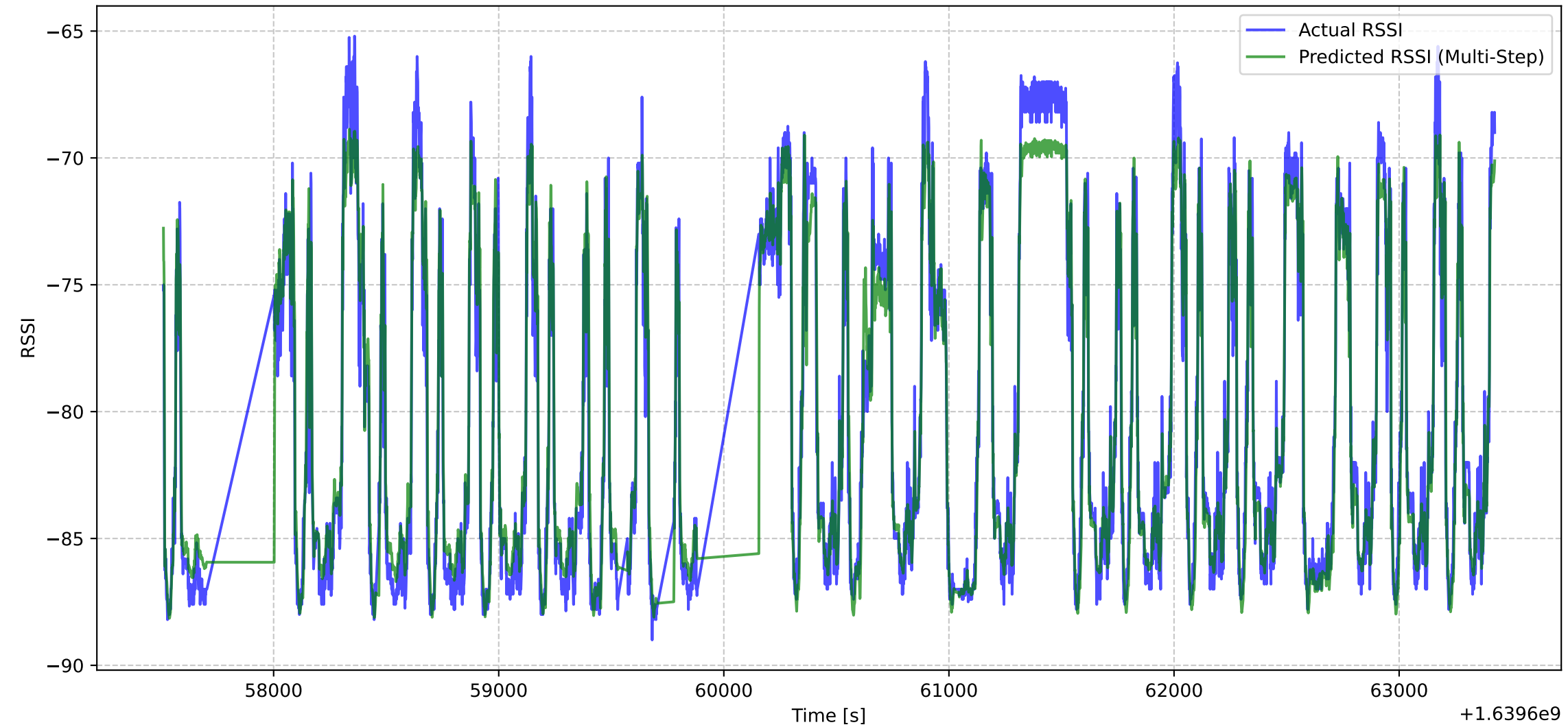
Predicted RSSI in Spatial Context (First Step)



Multi-Step Forecast Error by Step



Actual vs Predicted RSSI (First Step vs Multi-Step)



Resumo Automático:

- Forecast Steps configurados: 10
- Passos com MAE ≤ 2.0 : 10 passo(s)
- MAE no último passo: 0.0688

□ O modelo mantém boa acurácia durante quase toda a sequência de previsões.